Revisiting Neutrinos via Charm Decays in Astrophysical Sources

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Abstract

A recheck of the code behind my bachelor thesis found a unit error in the conversion of cross sections, a mismatched normalization of the charm cross section and an unbounded optical depth, and a vectorized rebuild with improved integration runs about sixty times faster. With these corrected, magnetar results hold and charm separates more clearly from pions and kaons, while charm in the accretion disk takes over at much higher energies than stated in the thesis.

1 Corrections

The thesis code was rebuilt as one vectorized program. It reproduces charm spectra to 10^{-6} relative precision using the thesis settings, while every parametrization agrees with its source and the method itself stands.

Units: Cooling factor and optical depth convert cross sections from millibarn to square centimetres with the factor that belongs to barn. Each target therefore acted a thousand times denser than stated.

Charm: In the thesis code, the charm cross section was divided by an extra factor of roughly 22 after its integral over the entire energy fraction range was compared to data from forward measurements (Aaij 2016). Most of that integral however comes from energy fractions far below the measured window, and inside the window the parametrization (Gonçalves 2007) already gives 1.07 times the data, so no extra factor was needed.

Depth: The optical depth multiplied spectra without bound, although the share of interacting protons cannot exceed unity. The corrected model uses the interaction probability instead, matching the thesis variant without optical depth as long as the ejecta stay opaque.

Numerics: Trapezoid weights replace the first order sum, a stable exponential removes a cancellation in the cooling factor and the vectorized integration runs about sixty times faster.

2 Magnetar

All figures compare the thesis on the left with the corrected results on the right and are normalized to the charm peak, shaded bands mark a factor of three around charm, and the reference (Carpio 2020) given in **Figure 1** shows pions and kaons with and without muon decay. In the light curves depicted in **Figure 2** charm still peaks first followed by kaons and pions, but all cooling features move to earlier times by about one decade. Charm now peaks after 7.1×10^3 s shortly after the spindown time of 3.2×10^3 s as is described in the reference, instead of after 4.9×10^4 s in the thesis.

The charm composition in **Figure 4** holds, with D^0 and D^+ carrying shares of 0.51 and 0.38 at the maximum and the individual peaks ordered by lifetime.

In the fluence of **Figure 3** charm exceeds both meson components above 2.9×10^9 GeV and leads the larger one at 10^{10} GeV by a factor of 14.5 instead of 3.0 so that even the lower edge of its band clears pions and kaons. The discrepancy that the thesis attributed to the optical depth disappears with the unit correction, and the initial proton energy of the thesis remains the correct evaluation of the formula given in the reference.

3 Accretion Disk

The disk result changes. Still, the charm composition in **Figure 5** keeps its shape. However, at the stated density of $10^{14}\,\mathrm{cm}^{-3}$ pions instead of kaons dominate the low energies in **Figure 6** and charm takes over only above 1.6×10^{10} GeV instead of 3.7×10^8 GeV in the thesis. At 10^{11} GeV it leads by a factor of 2.3 instead of 6.8 so that its lower band edge stays below the kaons. As **Figure 7** shows, the thesis takeover energy returns only near $4.6 \times 10^{15}\,\mathrm{cm}^{-3}$ which lies more than one decade above the stated density.

4 Conclusion

Overall, the thesis method stands while the magnetar conclusions become stronger, with the disk conclusions holding only for a much denser target.

Aaij, R. et al. *JHEP* (2016)
Carpio, J. A. et al. *PRD* (2020)
Gonçalves, V. P. et al. *JHEP* (2007)

Magnetar Scenario

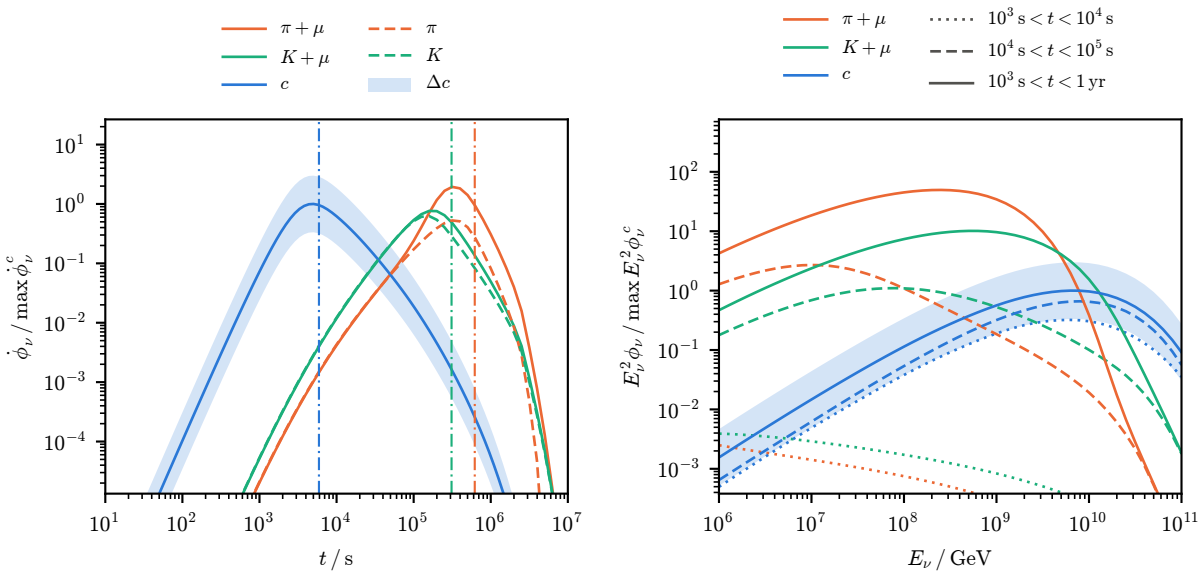


Figure 1: Reference light curve at $E_\nu = 10^9\,\mathrm{GeV}$ and fluence (Carpio 2020) normalized to the charm peak.

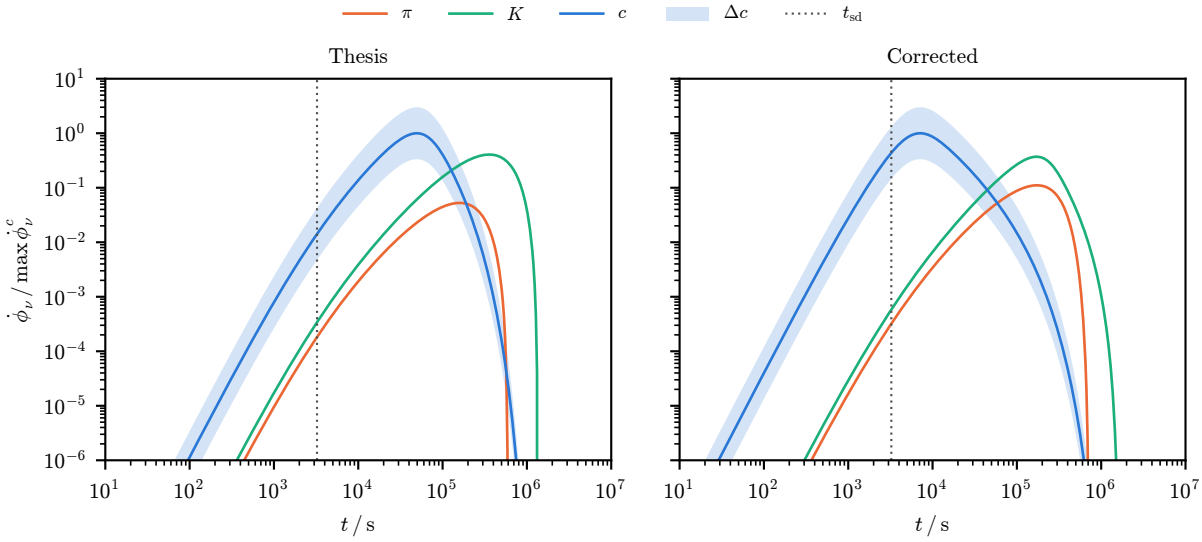


Figure 2: Light curves at $E_\nu = 10^9\,\mathrm{GeV}$ normalized to the charm maximum.

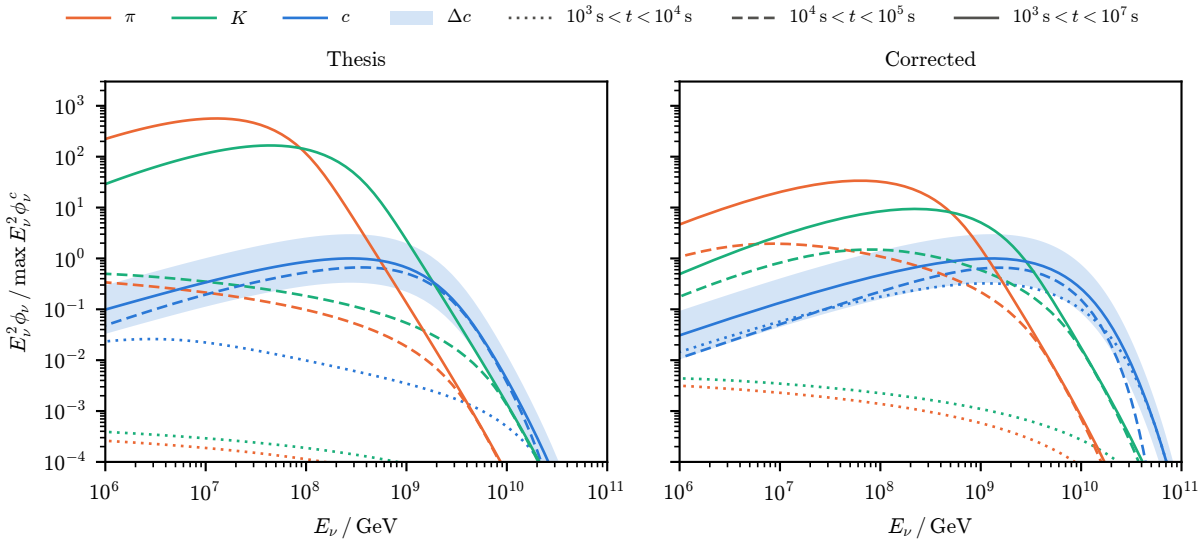


Figure 3: Fluence normalized to the charm peak of the widest time window.

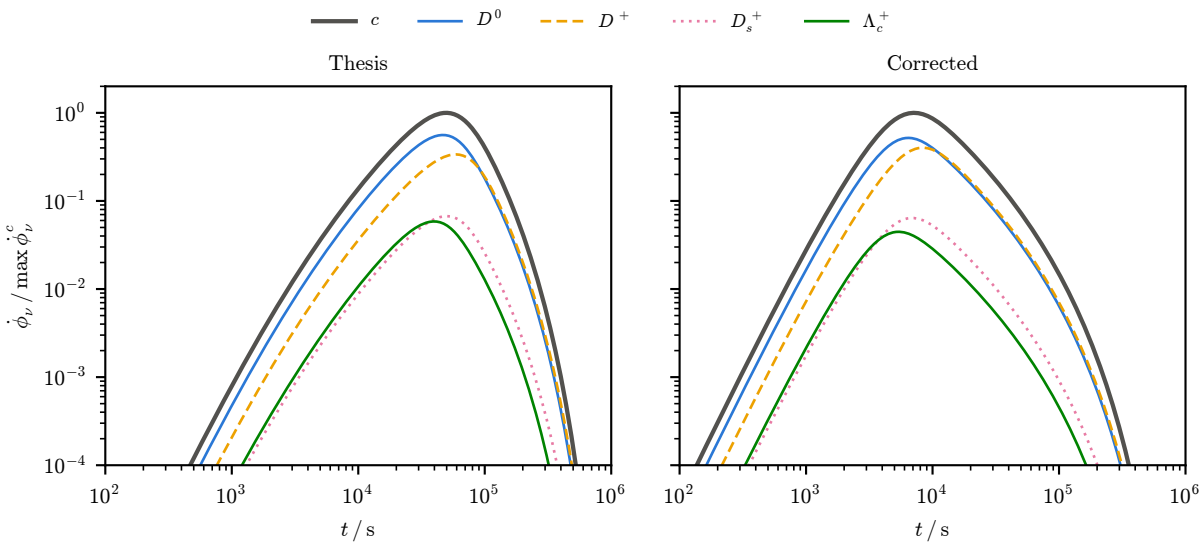


Figure 4: Charm hadrons in the light curve normalized to the total charm maximum.

Accretion Disk Scenario

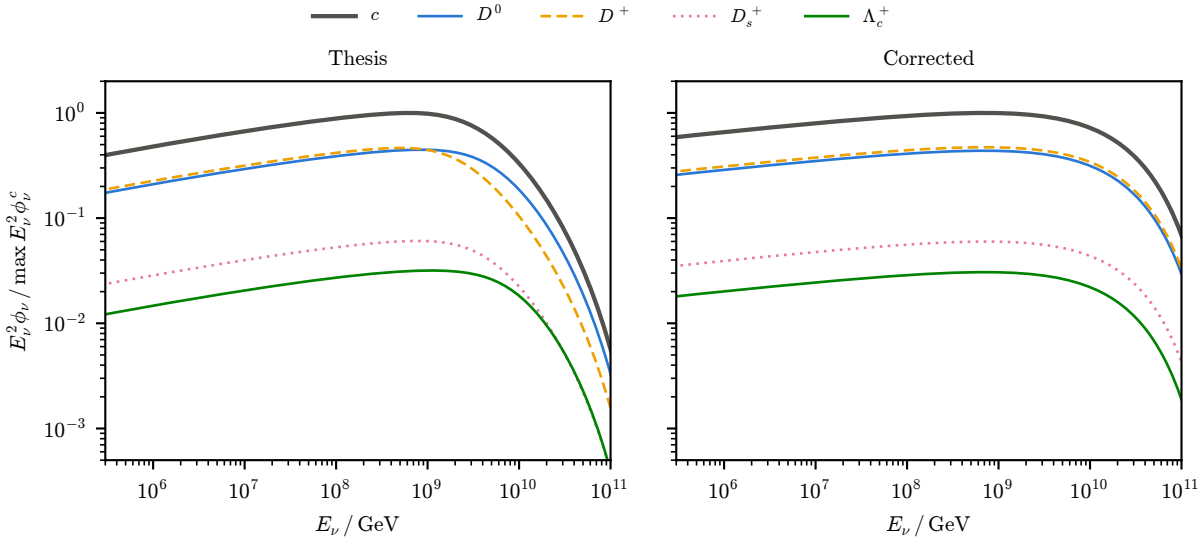


Figure 5: Charm hadrons in the disk spectrum normalized to the total charm maximum.

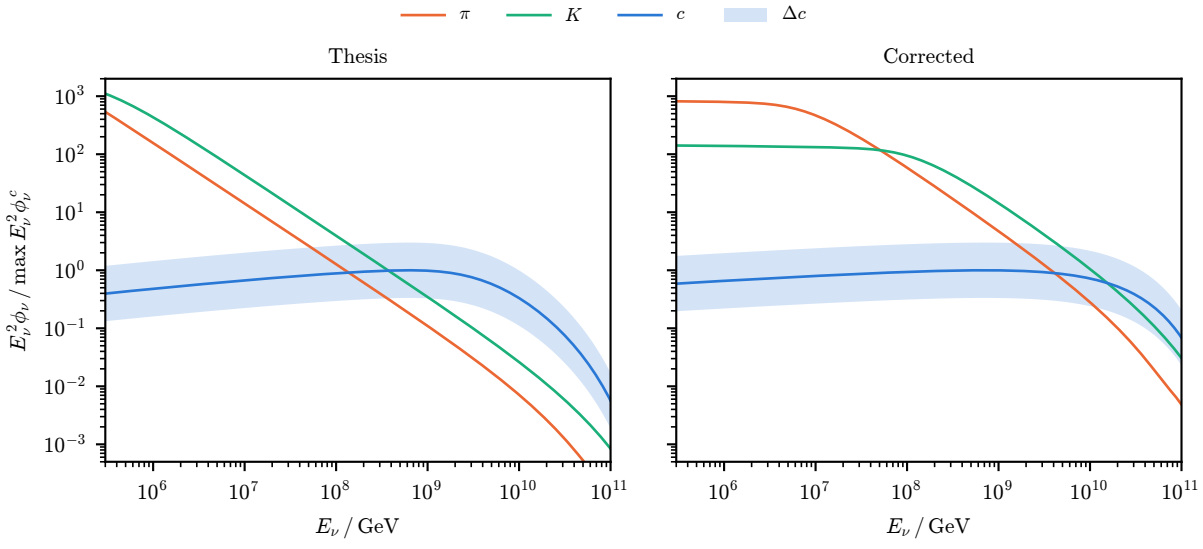


Figure 6: Disk spectra by parent normalized to the charm maximum.

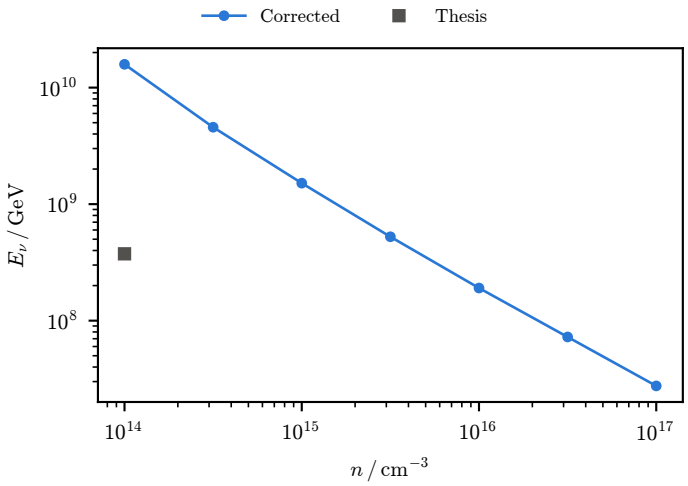


Figure 7: Neutrino energy at which charm overtakes pions and kaons in the disk against density after the corrections, compared with the thesis value at its stated density.

Declaration: The author acknowledges the use of artificial intelligence products by Anthropic, specifically of the models Claude 5.5 Sonnet and Claude 5.5 Opus for rechecking and rebuilding the code, preparing the figures and drafting this summary. The author independently verified all outcomes and maintains full accountability for this document.